

Yes. For your hackathon, make the PPT tell a problem → solution → technology → demo → comparison → impact story. Here is ready-to-use PPT matter for your Fake Website Detector Chrome Extension.

Slide 1 — Title

 WEBGUARD AI

AI-Powered Fake Website Detection & Cyber Awareness

A Chrome Extension for Real-Time Phishing Protection

Team: [Your Team Name]

Hackathon: [Hackathon Name]

Slide 2 — The Problem

 Why Fake Websites Are Dangerous

Every day, people use websites for:

Online banking and payments

Shopping

Social media

College and job applications

Email and cloud services

Attackers create websites that look almost identical to legitimate websites.

Human problems

Users may not notice a fake URL.

Fake login pages look genuine.

People often trust websites because they look professional.

Users may click links without checking the destination.

Traditional warnings can be confusing to non-technical users.

A single mistake can expose passwords or personal information.

Key problem:

> Humans are often the weakest link because phishing attacks target human decisions, not just technology.

Slide 3 — Our Solution

 WebGuard AI

A Chrome Extension that analyzes websites while the user is browsing.

It provides:

-  URL Analysis
-  AI-Based Risk Analysis
-  Threat Intelligence
-  Risk Score
-  Real-Time Warning
-  Awareness Explanation

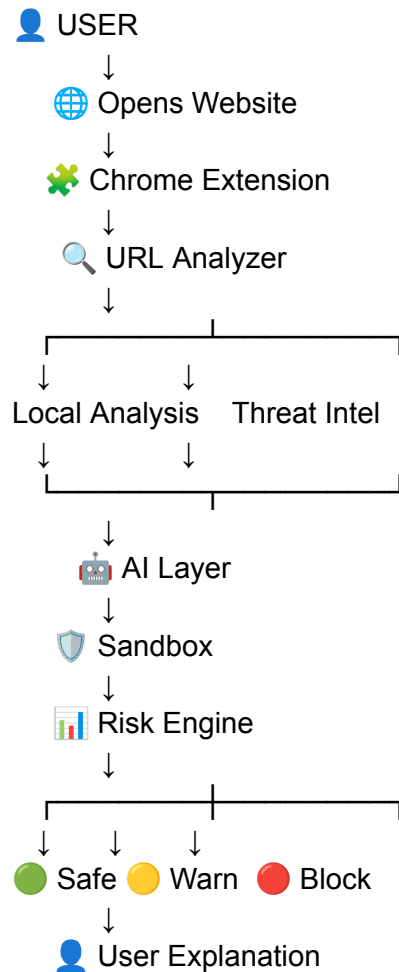
Instead of simply saying:

>  "Website is dangerous"

our system explains:

> Why the website appears suspicious and what the user should do.

Slide 4 — How Our System Works



Slide 5 — URL Analysis

What We Check

Our first layer analyzes the website URL.

Examples of suspicious indicators:

Look-alike domains

Suspicious keywords

Very long URLs

Excessive subdomains

IP address instead of domain name

Suspicious characters

Punycode domains

Unusual redirects

HTTP/HTTPS information

Example

Legitimate:
amazon.in

Suspicious:
amazon-login-security-example.com

The system doesn't automatically call something dangerous based on one indicator. It combines multiple signals.

Slide 6 — AI Detection 

Why AI?

Attackers continuously create new phishing websites.

A fixed blacklist cannot know every newly created malicious website.

Our AI layer can analyze multiple signals and produce:

Risk Score

0 – 30  LOW RISK
31 – 70  SUSPICIOUS
71 – 100  HIGH RISK

AI explains the decision

Example:

>  High Risk — 86/100

Reasons:

Domain resembles a known brand.

URL contains suspicious login-related terms.

Multiple unusual URL components detected.

Website behavior requires further inspection.

Slide 7 —  Sandbox Technology

Why Sandbox?

Some websites cannot be judged only from their URL.

A suspicious page may contain:

JavaScript behavior

Redirects

Hidden forms

Suspicious scripts

Unexpected downloads

Credential-collection behavior

Instead of allowing potentially dangerous content to interact directly with the user's environment, we can analyze suspicious content in an isolated sandbox environment.

Suspicious Website



Isolated Sandbox



Behavior Analysis



Suspicious Activity?



AI + Risk Engine

Important

> Sandboxing adds an additional analysis layer without exposing the user's normal browsing environment to the suspicious content.

For your hackathon, you can present sandbox analysis as a controlled backend/analysis environment, rather than claiming the Chrome extension itself is a complete malware sandbox.

Slide 8 — Chrome Extension 🧩

Why a Browser Extension?

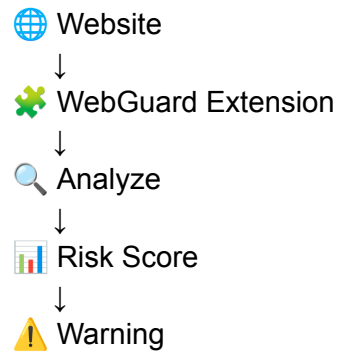
The user doesn't need to:

- ❌ Copy the URL
- ❌ Open another website
- ❌ Install a separate application
- ❌ Understand cybersecurity

Instead:

Browse → Detect → Explain

The extension works directly inside Chrome.



Slide 9 — Real-Time Protection

Example User Experience

User clicks a suspicious link.

WebGuard immediately shows:

 HIGH RISK

Risk Score: 86/100

 Possible phishing website

Why?

- Suspicious domain similarity
- Login-related keywords
- Unusual URL structure
- Threat intelligence warning

 Don't enter passwords

 Don't enter payment details

[View Explanation]

This makes the system useful even for non-technical users.

Slide 10 — Comparison With Existing Approaches

Feature	Basic Browser Warning	Blacklist	URL Checker Website
WebGuard AI			
Real-time browser protection	✓	✓	✗
URL analysis	Limited ✗	✓	✓
New/unknown sites	Limited ✗	Depends	✓
AI analysis	✗	Sometimes	✓
Behavior analysis	Limited ✗	Limited	✓
Explain why risky	Limited ✗	Limited	✓
Sandbox analysis	✗	Rare	✓
User awareness	Limited ✗	Limited	✓
Chrome integration	✓	✗	✓

Our advantage

> We combine browser-level convenience, URL analysis, threat intelligence, AI reasoning, isolated behavior analysis, and user education in one workflow.

Slide 11 — What Makes WebGuard Different?

🧠 "Detect + Explain + Educate"

Most security tools focus on:

> "Is this website safe?"

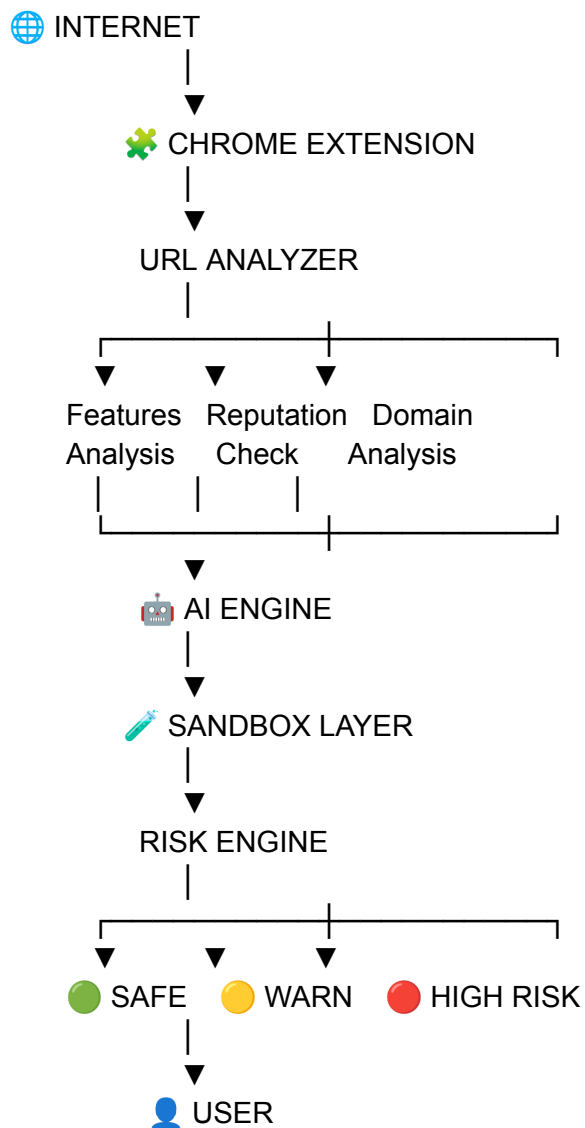
Our project focuses on:

> "Is this website safe, WHY is it suspicious, and WHAT should the user do?"

Three-layer protection

- 1 Detection
Find suspicious indicators.
- 2 Intelligence
Compare against reputation/threat information.
- 3 Explanation
AI converts technical findings into simple language.

Slide 12 — System Architecture



Slide 13 — Technology Stack

Frontend

HTML

CSS

JavaScript

Chrome Extension Manifest V3

Security

URL Feature Extraction

Domain Analysis

Threat Intelligence

Reputation Checking

Suspicious Behavior Analysis

Sandbox Environment

AI

AI-based classification

Risk explanation

Natural-language security recommendations

Optional Backend

Node.js / Python

REST API

Database for detection logs

Slide 14 — Hackathon Demo



Live Demonstration

Step 1: Install WebGuard extension.

Step 2: Open a safe website.



LOW RISK

Step 3: Open a deliberately created demo phishing website in the controlled hackathon environment.

Step 4: Extension analyzes the URL.

Step 5: AI analyzes the collected indicators.

Step 6: Sandbox performs controlled behavior analysis.

Step 7: Extension displays:



HIGH RISK

Risk Score: 89/100

Reasons:



Suspicious domain



Login impersonation



Unusual URL structure



Suspicious behavior

Recommendation:

Do not enter sensitive information.

Slide 15 — Impact

Who Can Use It?

 Students

 Employees

 Families

 Online banking users

 Online shoppers

 Organizations

Benefits

Reduces phishing risk

Helps users make better decisions

Provides understandable security information

Detects suspicious websites in real time

Improves cyber awareness

Adds multiple layers of protection

Slide 16 — Future Scope

Future Improvements

Mobile browser support

Multilingual warnings

More advanced AI models

Continuous threat-intelligence updates

QR-code URL scanning

Email phishing detection

SMS/link analysis

Enterprise dashboard

Adaptive detection based on new attack patterns

Slide 17 — Final Message

 WebGuard AI

"Don't just detect the threat. Understand it."

Detect  → Analyze  → Isolate  → Explain  → Protect 

> Our goal is not only to identify fake websites, but to help people understand phishing and make safer decisions online.

Thank You 🙏

Questions?